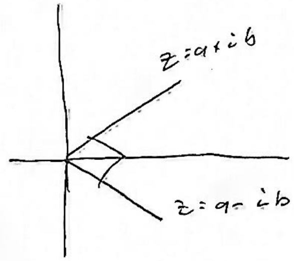
<u>Recall:</u> $z = a + ib$ (2d deg. free)

matrix $m \times n$ has $2mn$



Real $u, v \in \mathbb{R}^d$ $\langle u, v \rangle = \sum_{j=1}^{d} u_j v_j = u^T v$ symmetric + bilinear

Complx $u, v \in \mathbb{C}^d$ $\langle u, v \rangle = \sum_{j=1}^{d} \overline{u_j} v_j = u^+ v$, $u^+ = \bar{u}^T$

conj. symm $\langle u, v \rangle = \overline{\langle v, u \rangle}$

sesquilinear conjugate linear in first, linear in second

L layers
H attn heads
d embedding dim
V vocab size
T max seq len
vects $\in \mathbb{C}^d$
matr $\in \mathbb{C}^{m \times n}$
$z = a + bi$ 2 real dof
compl vect of dim d ~~converts~~ 2d real dof

Loss is <u>not</u> holomorphic $L: \mathbb{C}^n \to \mathbb{R}$ (real valued), need Wirtinger calculus

$$\partial_z = \frac{1}{2}\left(\partial_a - i\partial_b\right)$$

$$\partial_{\bar{z}} = \frac{1}{2}\left(\partial_a + i\partial_b\right)$$

Noor et al (2023) Wirt Calc
Kreutz-Delgado (2009) Complex gradient

Steepest ascent direction of L_{real} wrt to z is

$$\partial_a L + i\partial_b L = 2\partial_{\bar{z}} L$$

PyTorch $g = \partial_a L + i\partial_b L$ as gradient

with update $z \leftarrow z - \eta g$

$$z' = (z - z_0)\, e^{i\theta} + z_0$$

(scaling; angle; anchor point)

$\bar{z} = z$ in Real case, so Wirt is just calcul

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Token Embedding

stored in $W_E \in \mathbb{R}^{V \times 2d}$ (vocab), reshaped to complex on forward pass

given $t \in \{0, \ldots, V-1\}$

$$r_t = W_E[t, :] \in \mathbb{R}^{2d}$$

$$e_t = \underbrace{\texttt{view\_as\_complex}(r_t)}_{\text{PyTorch pragmatism, no native } \mathbb{C} \text{ support}} \in \mathbb{C}^d$$

not native complex embedding $e_t = W_E[t, :] \in \mathbb{C}^d$

↳ j^{th} is $e_{t,j} = r_{t,2j} + i r_{t,2j+1}$ adjacent dimension pairing where consecutive pairs of real vals become real & imaginary

→ see Norm!

↳ without it, amplitude encoding requires norm step at evaluation time, introducing nonlinearity not present in training

isometric isomorphism $\mathbb{R}^{2d} \to \mathbb{C}^d$

$$\|r\|^2_{\mathbb{R}^{2d}} = \sum_{k=0}^{2d-1} r_k^2 = \sum_{j=0}^{d-1} \left(r_{2j}^2 + r_{2j+1}^2\right) = \sum_{j=0}^{d-1} |e_j|^2 = \|e\|^2_{\mathbb{C}^d}$$

So, W has $2Vd$ real params, $\partial_{\bar{e}_t} L = \partial_{r_t} L$

Not native $\mathbb{C}^{V \times d}$? Suppose we could via complex Gaussian entries (normalized?).

we use nn.Embedding$(V, 2d)$ (with default initialization uniform on $[-1,1]/\sqrt{V}$ or normal (depends on version))

no 2^n constraint

3 no $d = 2^n$ where embeddings load directly to n qubits. we default to $d = 256 = 2^8$

init differs [1. uses Pytorch default applied in $\mathbb{R}^{2d}$

no unit norm constraints [2. No embedding onto unit sphere, so are unconstrained in $\mathbb{C}^d$, (no $\|e\| = 1$) → 2

$|\langle e_a, e_b \rangle|^2 \notin [0, 1]$, $\text{sim}_{\text{learn}} \approx \text{sim}_{\text{meas}} (C^{(m)}_{ab})^2$ not hold

Position Encoding

- we use fixed, non learned complex sinusoidal encoding

pos $i \in \{0, \dots, T-1\}$ dim $j \in \{0, \dots, d-1\}$

angular freq $\omega_j = \frac{1}{10000^{j/d}}$ and angle $\phi_{ij} = \omega_j \cdot i$] $\omega_j \downarrow$ from $\omega_0 = 1$ to $\omega_{d-1} = \frac{1}{10000}$

↳ low indexed dims oscillate rapidly, slowly high ind.

every dim gets sin + i cos

trad is even sin odd cos

↳ different inductive bias

$PE_{ij} = \sin(\phi_{ij}) + i\cos(\phi_{ij})$, $\sin\theta + i\cos\theta = -i(\cos\theta + i\sin\theta)$

$= -i e^{i\theta}$

$= -i e^{i\phi_{ij}}$

$= -i e^{i\omega_j \cdot i}$ (pure phase rotation as position)

up to global factor of $-i$, constant

↳ then $e_t + PE_i + s$

$|PE_{ij}| \overset{!}{=} 1$

↳ pos emb contributes phase, not magnitude to $e_t + PE + S$

↳ code uses add, so x not unit magnitude

Same as trad, but every dim gets complex exponential

- Non Learned?

compare $W_p \in \mathbb{R}^{T \times d}$ pos, input to first layer

add $W_s \in \mathbb{R}^{2 \times d}$ segment

$x_i^{(0)} = e_{x_i} + W_p[i,:] + W_s[seg_i,:] \in \mathbb{R}^d$

↳ backward pass distributes upstream grad

add $W_p \in \mathbb{C}^{T \times d}$, $W_s \in \mathbb{C}^{2 \times d}$, x_i ---- componentwise addition

- mult $x_{ij}^{(0)} = (e_{x_i,j} + s_{seg_i,j}) \cdot e^{i\omega_j \cdot i}$ each component rotated in $\mathbb{C}$ (diagonal phase gate in Q)

backward pass undoes rot

$\partial_e L = \partial_x L \cdot e^{-i \dots}$

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